Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #23 | Smart Cities, Transport & Logistics

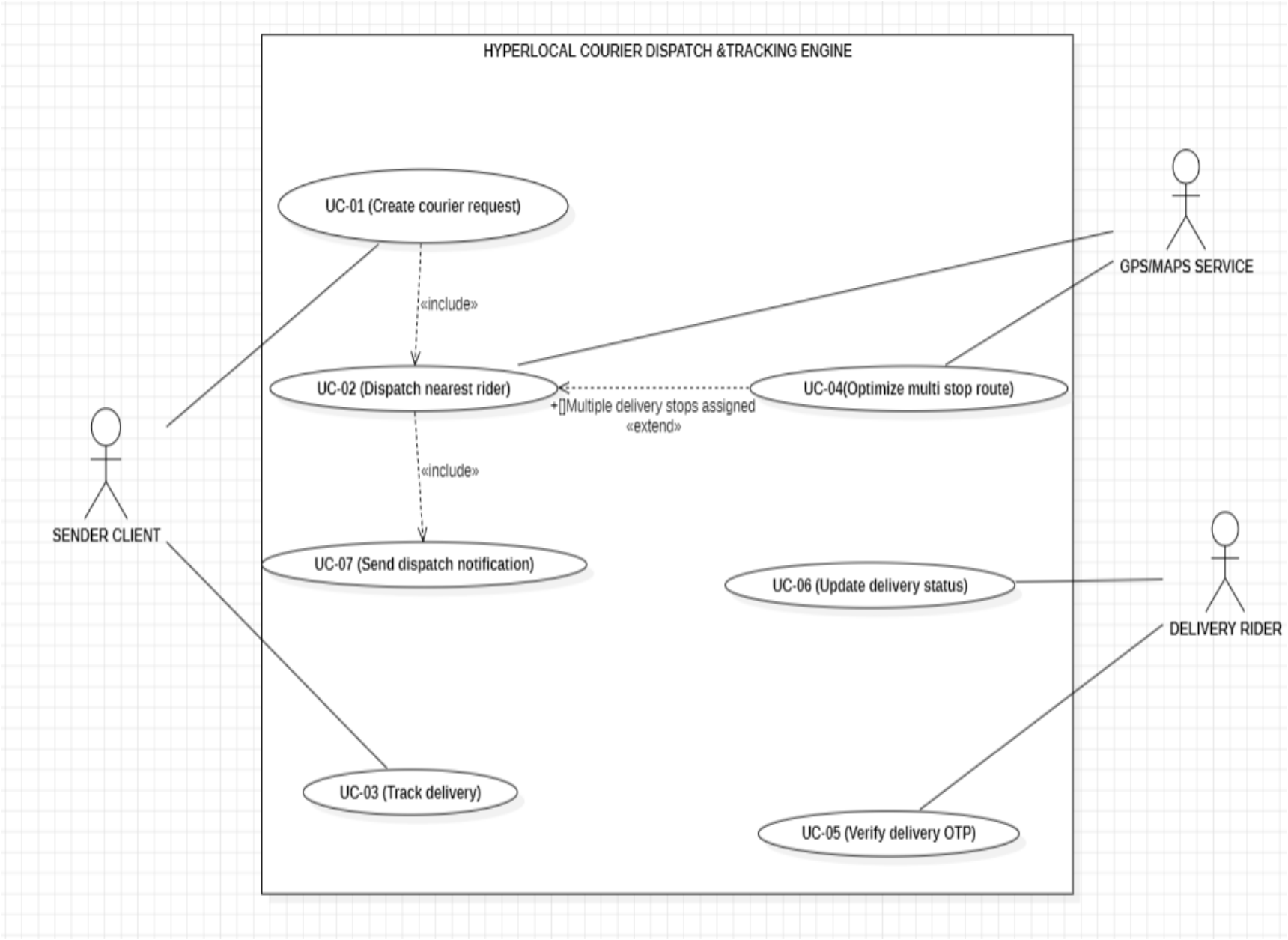
Hyperlocal Courier Dispatch & Tracking Engine

REQIREMENT TABLE:

Req ID	TYPE	DESCRIPTION	PRIORITY	ACCECPTANCE CRITERIA	RATIONALE
FR-001	FUCNTIONAL	The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius.	HIGH	PASS: An active rider within 3 km receives the dispatch notification. FAIL: The order is assigned to an offline rider or an ineligible rider when an eligible rider exists.	Core dispatch functionality of the courier platform.
FR-002	FUCNTIONAL	The system shall allow the sender to create a courier request by entering pickup location, destination location, and parcel details.	High	PASS: A courier request is created when all mandatory information is valid. FAIL: The system accepts a request when a required location or parcel detail is missing.	A courier request is required before the system can dispatch a rider
FR-003	FUCNTIONAL	The system shall display the assigned rider's current location and delivery status to the sender client while the courier is active.	HIGH	PASS: The sender client displays updated rider location and delivery status while the rider is active. FAIL: The sender cannot view the rider's location or delivery status during an active delivery.	Provides the tracking functionality required by the courier platform.
FR-004	FUCNTIONAL	The system shall generate an optimized route for a delivery rider when multiple active	MEDIUM	PASS: When two or more delivery stops exist, the generated route contains every assigned stop. FAIL: One or more assigned stops are omitted from the generated route.	Supports the multi-stop routing functionality specified in the problem statement.

		delivery stops are assigned.			
FR-005	FUCNTIONAL	The system shall verify the destination OTP entered by the delivery rider before marking a courier order as successfully delivered.	HIGH	PASS: A correct OTP changes the delivery status to "Delivered." FAIL: An incorrect OTP does not complete the delivery.	Provides the destination verification required by the system.
NFR-001	NON-FUCNTIONA-PERFORMANCE AND SECURITY	Delivery status and live GPS telemetry updates from riders shall transmit to the sender client with under 2-second latency.	HIGH	PASS: Benchmarking confirms that delivery status and GPS updates reach the sender client in under 2 seconds. FAIL: The measured latency is 2 seconds or more.	Ensures near-real-time delivery tracking for the sender.
NFR-002	NON-FUCNTIONAL-SECURITY	The system shall encrypt sensitive delivery and OTP-related data during transmission.	HIGH	PASS: Security testing confirms that sensitive delivery and OTP-related data is transmitted through encrypted connections. FAIL: Sensitive data is transmitted without encryption.	Protects sensitive delivery and verification information during communication.

UML Use-Case Diagram:



Use-Case Flow Document:

UC-02 — Dispatch Nearest Rider

Main Success Scenario

1. The Sender Client submits a valid courier request containing the pickup location, destination location, and parcel details.

2.The system validates the courier request & identifies active delivery riders near the pickup location.

3. The system obtains the current location of the eligible riders.

4. The system filters the riders to those within the required 3 km radius.

5. The system identifies the nearest eligible active delivery rider & assigns the courier request to the selected rider.

6. The system sends a dispatch notification to the assigned rider.

7. The rider receives the dispatch notification.

8. The system updates the courier request status to **Assigned**.

9. The Sender Client can view the assigned delivery and track its status.

10. The use case ends successfully.

Alternate Flow — No Eligible Rider Available

1. At Step 5 of the Main Success Scenario, the system determines that no active rider is available within the required 3 km radius.

2. The system does not assign a rider to the courier upon request.

3. The system keeps the courier request in an **Unassigned/Pending** state.

4. The system informs the Sender Client that no eligible rider is currently available.

5. The use case ends without assigning a rider.